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The impact of refrigeration on food losses and associated greenhouse gas emissions throughout the supply chain

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Abstract

One-third of food produced globally is wasted while approximately 800 million people suffer from hunger. Meanwhile, food losses produce approximately 8% of total anthropogenic greenhouse gas (GHG) emissions. This study develops a food loss estimation tool to assess how improved access to the cold chain could impact food loss and its associated GHG emissions for seven food types in seven regions. This study estimates that poor cold chain infrastructure could be responsible for up to 620 million metric tons (Mmt) of food loss, responsible for 1.8 $GtCO_2$-eq annually. Utilizing fully optimized cold chains could save over 100 Mmt of fruit and vegetable loss in South & Southeast Asia and over 700 Mmt CO2-eq in Sub-Saharan Africa. Developing more localized, less industrialized ('farm-to-table') food supply chains in both industrialized and non-industrialized contexts may save greater quantities of food than optimized cold chains. Utilizing localized supply chains could save over 250 Mmt of roots and tubers globally (over 100 Mmt more savings than those of an optimized cold chain) and reduce GHG emissions from meat losses in industrialized regions by over 300 Mmt CO2-eq. Due to the differences in the environmental intensity of food types, cold chain investments that prioritize reducing overall food losses will have very different outcomes than those that prioritize reducing GHG emissions.

1. Introduction

This study uses a food loss estimation tool to quantify changes in food loss and associated greenhouse gas (GHG) emissions that may occur with the introduction or quality improvements of cold chain technology, as well as the length of a food supply chain (FSC). The analyzed scenarios illustrate the differences between more localized, less industrialized FSCs and globalized, more technologically-advanced FSCs. By modeling food losses at each stage of the supply chain, this study highlights where the cold chain can be strategically deployed and optimized to direct food system investments to reduce food losses and emissions.

1.1. Refrigerated supply chain ('cold chain')

Ideally, the cold chain provides an unbroken, controlled atmospheric environment to ensure the quality and safety of perishable products throughout all stages of a supply chain (Ma and Guan 2009, Aung and Chang 2014). The 'cold chain' refers to both temperature and humidity control, incorporating both physical technology and logistical management (Garnett 2007, Heard and Miller 2019). In the context of this paper, the term 'refrigeration' is used to represent the suite of cold chain interventions, which vary according to the requirements of different food types and can include cool storage, frozen storage, and humidity control with or without temperature control. With regard to food supply, the cold chain extends from farms and processing plants to retail (grocery) and foodservice operations (Garnett 2011, Kitinoja 2013). The cold chain provides many safety, nutritional, and health benefits. By extending the shelf life of food, the cold chain can improve and expand access to perishable foods and reduce spoilage and foodborne illness (Heard and Miller 2019).

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The cold chain is also necessary for effective vaccine and antibiotic delivery (Heard and Miller 2019). A continuous, unbroken cold chain is necessary to maximize the benefits of safety and reduce product losses; however, in many non-industrialized economies[1], cold chain elements may have inconsistent quality, continuity or lack cold chain elements entirely (Ishangulyyev *et al* 2019).

This study analyzes the effects of moving from the current state of inconsistent and variable quality cold chains throughout the world to an optimized system.

1.2. Broad impacts of food loss and waste

The United Nations' Sustainable Development Goals 2 and 12 mention achieving food security and improved nutrition, and addressing food losses along supply chains, respectively (UN 2015). Delivering on these goals is critical from humanitarian, environmental, and financial perspectives and the cold chain can have a role in achieving these objectives. While estimates vary, approximately one-third of food produced globally, 1.3 Gt, is wasted, equating to approximately 4.4 GtCO2e annually (Gustavsson *et al* 2011, FAO 2015). Concomitantly, it is estimated that 720–811 million people suffer from hunger (FAO *et al* 2020). The financial cost of food loss and waste alone (excluding fish & seafood) is $750 billion annually; this does not take into account the financial costs of disposal, logistics, environmental damage, nor the human potential lost if the food were effectively distributed (FAO 2013).

Understanding where losses occur in the FSC is critical to addressing systemic inefficiencies that contribute to both hunger and climate change. While food loss and waste are global issues, the patterns of food loss and waste differ. In higher income, more industrialized regions, a greater proportion of food is wasted (>40%) at the consumption phase of the FSC. In lower income regions, more than 40% of food losses occur in the early stages (post-harvest and processing) of the FSC, often due to poor logistics and lack of climate control via the cold chain (Gustavsson *et al* 2011). While fully-developed and under-developed cold chains are often represented as binary, mutually distinct states, the reality is that development of a cold chain is a stepwise, context-specific process. This model examines differences in refrigeration qualities (none, poor, average, good) for each stage of a FSC, as well as a comparison of long multi-stage FSCs with very short farm-to-consumer FSCs. This model can provide critical insights into the region- and food type-specific tradeoffs of cold chain implementation, thereby informing optimal FSC development.

1.3. Prior FSC-cold chain research

Research on food systems and the cold chain has been growing over the past couple decades but remains fragmented and limited in terms of direct applicability to FSC stakeholders. Most studies fall into one of two types: historically based (meta) analysis and theoretical projection models. The former uses historical data to assess trends and rationalize those trends on regional and global scales. Gustavsson *et al* (2011) presents a critical meta-analysis of global FSCs, examining the stage-specific losses regionally and providing insights into the causes behind regional food losses and potential solutions. Porter *et al* (2016) built upon Gustavsson's research to publish region- and food-specific emission factors for various food products and food types. The latter approach has largely utilized storage conditions, namely temperature and time, to model food degradation and loss. James and James (2010) provide one of the earlier and more robust analyses on the relationship between food loss and climate change. Several other studies have focused on cold chain development in China, utilizing conditions-based frameworks to illustrate how various environmental factors can impact shelf life and food loss (Dong and Miller 2021, Hu *et al* 2019, Wu *et al* 2022).

More recent academic and industry studies have used historical data and development trends to model and understand how refrigeration may manifest in non-industrialized regions. Heard and Miller (2019) model the development of a Sub-Saharan African cold chain, including food losses, dietary changes, and the emissions associated with those factors. The Global Food Cold Chain Council (2015) modeled cold chain development by comparing cold chain penetration and food loss and waste data in non-industrialized regions with that of industrialized regions.

Unlike previous research, this study focuses on potential improvements that can be realized by cold chain upgrade and optimization at specific stages within the FSC, focusing specifically on partial or suboptimally functioning refrigeration. Additionally, this study explicitly compares the losses and associated emissions of shorter, less refrigerated FSCs with extended, more refrigerated FSCs.

2. Methods

The model's scope and components were defined, including the FSC stages, refrigeration qualities and their associated loss rates, and relevant emissions

[1] There is a range of language used across authors and disciplines to describe varying levels of development—developed vs. developing countries/economies, low- and middle-income countries vs. high income countries, non-industrialized vs. industrialized countries. We have opted to use non-industrialized vs. industrialized economies since our focus is the degree to which regional economies have industrialized their food supply chains.

factors tied to those losses. FAOSTAT Food Balance data were input into the model to test its efficacy and identify opportunities for regional FSC optimization. The full model is available in the supporting information and includes the option to customize FSC length, food types, regions, and loss rates to simulate a range of scenarios in addition to the results reported in this manuscript.

2.1. Defining the model's scope and components

While there is not universal consensus on the definition of food loss, it is generally understood that food losses are the quantitative and qualitative post-harvest decreases in food fit for human consumption (Chaboud and Daviron 2017). Solutions to food loss generally focus on management or technological changes to the FSC. In contrast, the term food waste encapsulates edible food that is supplied to the consumer but is never actually eaten, with solutions focusing more on behavioral shifts (Parfitt *et al* 2010, Gustavsson *et al* 2011, FAO 2019, Heard and Miller 2019, Dong and Miller 2021). This study focuses on food losses in the post-harvest to retail stages of the FSC (illustrated by the area within the red dotted line in figure 1). Agricultural Production and Consumption are included for context and to show the relative contribution of these stages, but interventions to reduce food loss or waste at these stages fall outside the scope of this paper as they do not pertain to management of the cold chain. Additionally, only quantitative food loss and waste is considered in this model, as qualitative changes are not reported in the datasets utilized.

In contrast to previous studies which have included five (Gustavsson *et al* 2011, Heard and Miller 2019) or seven (FAO 2019) FSC stages, the model developed for this study is customizable and includes a maximum of ten stages. The larger number of stages does not imply a longer supply chain but provides a higher level of resolution to be able to model individual transportation and storage stages separately. In prior studies with five stages, transportation is generally embedded into other stages. In some more recent studies, transportation is represented as a single stage or data point (FAO 2019). By including and accounting for transportation throughout the FSC, this study allows for greater differentiation between the potential impact of refrigerated transportation from earlier versus later stages which is critical considering the trend of greater early-stage food loss in non-industrialized regions (Parfitt *et al* 2010, Aschemann-Witzel *et al* 2015).

2.2. Model construction

The food loss and GHG model was developed using Microsoft Office Excel. This model is based on the FSC stages and associated food loss rates defined by Gustavsson *et al* (2011) and expanded upon by Dong and Miller (2021), FAO (2019), Heard and Miller (2019).

Three scenarios were modeled for every region and food type: first, a 'baseline' scenario employing current loss rates across the FSC; second, an 'optimized' scenario employing minimum loss rates at each FSC stage; and third, a 'short FSC' scenario, employing current loss rates across a 4-stage supply chain (figure 1(b)). These scenarios were used to provide nuance with the understanding that within and between regional FSCs, there is a lot of variability in terms of their robustness—the length, duration, and presence of stages. These scenarios were modeled in two ways: first, using a standard food input quantity—100 000 kg (as shown in figure 2); second, using consumption data from FAOSTAT—due to data constraints, FSC losses were extrapolated from loss rates and applied to consumption as was done by Heard and Miller (2019) (Gustavsson *et al* 2011, FAOSTAT 2023). Ultimately, the differences in food loss and associated emissions were calculated between these scenarios, highlighting how and where refrigeration can have the greatest impact within the FSC.

2.3. Regions and food types

This study focuses on regions—Europe including Russia, Industrialized Asia, Latin America, North Africa & Central Asia, North America & Oceania, South & Southeast Asia, and Sub-Saharan Africa—and food types—cereals, fish & seafood, fruits & vegetables, meat, milk products, oilseeds & pulses, roots & tubers—as defined in Gustavsson *et al* (2011), Heard and Miller (2019), Porter *et al* (2016).

2.4. Food loss rates

Food loss rates at any FSC stage are influenced by a variety of factors including infrastructural, societal, logistical, behavioral, and environmental factors. Due to the complex interplay of these factors, the refrigeration quality estimates incorporate the entire range of attributes that lead to either poor or good food loss rates and do not separate out cold chain technology specifically.

2.5. Refrigeration quality

Four refrigeration quality levels were developed for each FSC stage in the future state scenario to balance the simplicity and utility of the study. Those levels include good, average, poor, and no refrigeration, and were defined using baseline food loss rates. Good refrigeration reflects the lowest baseline loss rate, while no refrigeration reflects the highest baseline loss rate. Average and poor refrigeration are defined by the equations below:

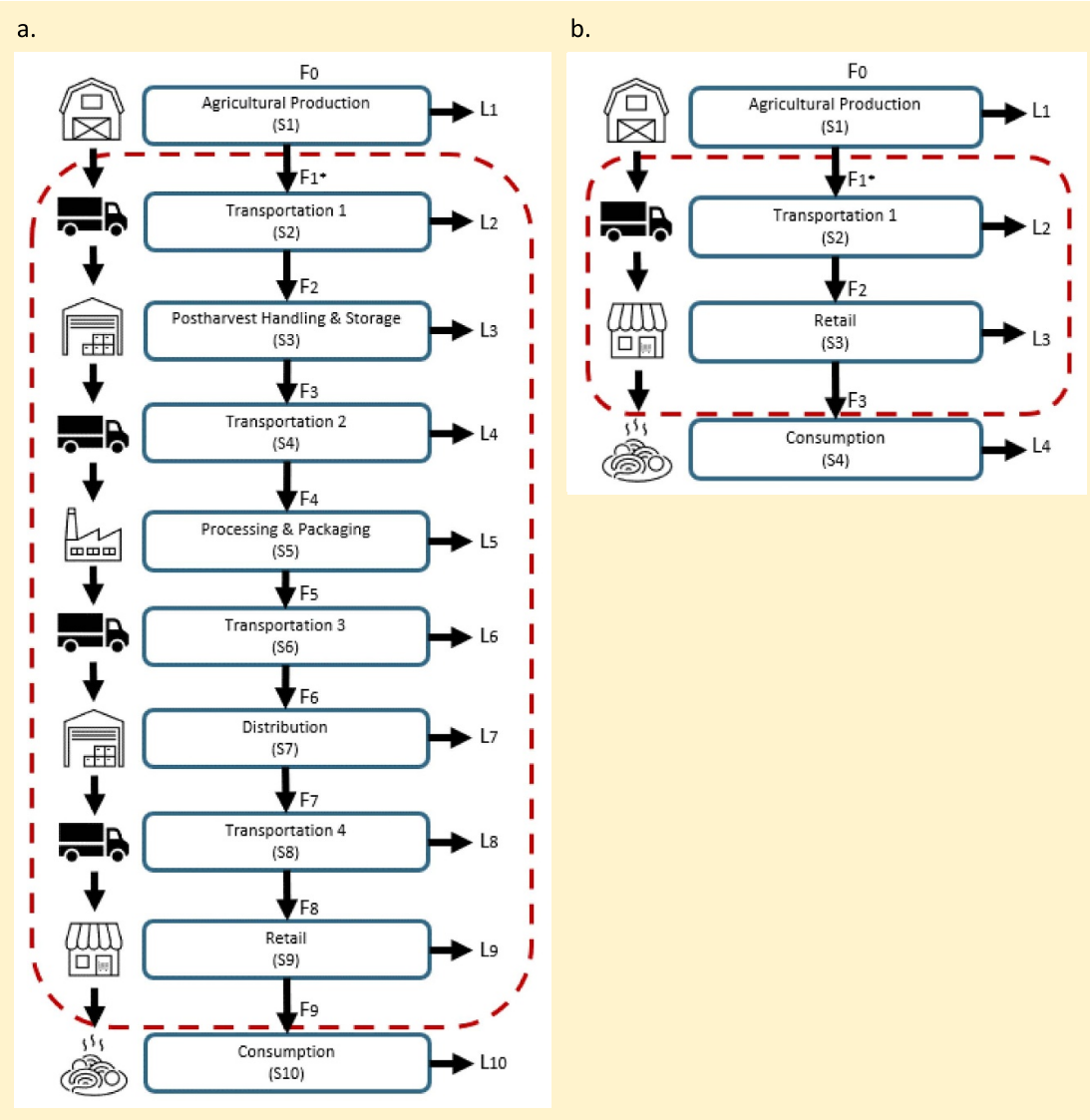


Figure 1. Visual representation of 10-stage FSC and 4-stage FSC and the mass flows for food (F) and losses (L) that can be included in the model developed through this study.
Notes: Figure 1 illustrates both (a), a 10-stage FSC; this was used to model all current and optimized FSC scenarios. (b), a 4-stage FSC; this was used to model the short FSC detailed in figure 5. In both figures, Sn represents the FSC stage, Fn the food input into that stage, and Ln the food loss occurring in that stage. The red boundary indicates FSC stages that are directly impacted by refrigeration.

Average Refrigeration:

$$\text{Good Refrigeration \% food loss} + \frac{1}{3}(\text{max loss \%} - \text{min loss \%})$$

Poor Refrigeration:

$$\text{Good Refrigeration \% food loss} + \frac{2}{3}(\text{max loss \%} - \text{min loss \%}).$$

2.6. Baseline scenario

Baseline scenarios represent current levels of refrigeration with associated loss rates derived from Gustavsson *et al* (2011). To be able to model differences in refrigeration quality for transportation and storage phases, as well as distribution and retail, the 5-stage system (Gustavsson) was subdivided to a 10-stage system (figure 1).

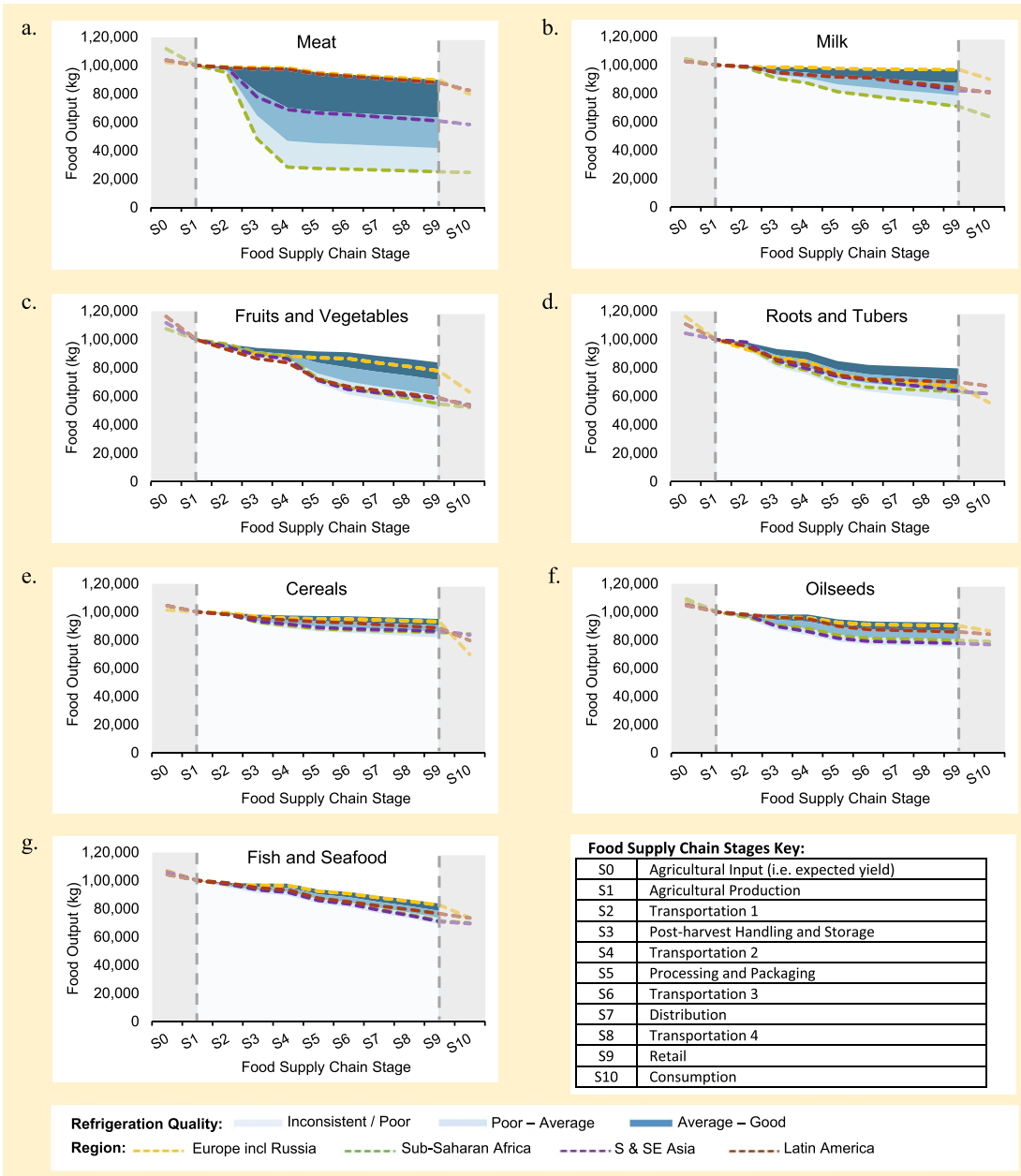


Figure 2. Food loss rates of different qualities of refrigeration for seven food types.
Notes: Figure 2 shows the current mass of food loss for 100 000 kg of each food type entering the FSC at different qualities of refrigeration. The blue shading zones represent expected food losses associated with inconsistent-to-poor, poor-to-average, and average-to-good refrigeration quality, with the upper end of the top band representing consistently good refrigeration, which is considered optimal in this model. To place refrigeration quality data into context, figure 2 also depicts current food losses associated with four regions that represent the spectrum of existing cold chain development—Europe: fully developed, Latin America and South & Southeast Asia: partially developing, and Sub-Saharan Africa: fully developing. While loss and waste in Agricultural Production (S1) and Consumption (S10) are displayed, they are not directly impacted by the cold chain.

2.7. Optimized scenario

Optimized scenarios represent a FSC with good-quality refrigeration across all stages by utilizing the lowest baseline loss rates for S1–S8.

2.8. Emission factors

Food loss emission factors (which allow for calculating kg CO2e from kg food loss) were sourced from Porter *et al* (2016) and Hamerschlag and Venkat (2011). These emissions factors are region- and food type-specific because different regions have different agricultural practices, inputs, energy sources, etc, producing different quantities of emissions. This model only accounts for emissions associated with food losses and does not include emissions resulting from refrigeration or other supply chain operations (e.g. transportation). For greater detail on the food types, associated foods within each food type, and their emissions, see section 4 of supplementary information (Heard and Miller 2019).

3. Results

The results indicate the greatest overall loss rate improvements from increased refrigeration would occur in Sub-Saharan Africa, North Africa & Central Asia, South & Southeast Asia, and Latin America, particularly with respect to meat, milk products, and fruits & vegetables.

4. The impact of refrigeration quality by stage and food type

The results highlight differences in how refrigeration quality affects food losses for different food types (see figure 2, which depicts food losses with respect to differences in refrigeration quality). While optimized refrigeration can reduce losses by 26%–63% for milk products, fruits & vegetables, and meat, it has a lesser impact, 13%–20% reduction, with respect to cereals, fish & seafood, oilseeds & pulses, and roots and tubers. This is reflected in the fact that regional differences in food losses are more pronounced for meat, milk products, and fruits & vegetables. Moreover, the results show nuances in terms of where within the FSC food losses occur, and consequently where refrigeration would be most effectively implemented. For instance, improved refrigeration implemented in post-harvest handling & storage (S2) is most effective for preventing meat loss (see figure 2(a)), while refrigeration implemented in processing & packaging (S4) is most effective for preventing fruit & vegetable loss (figure 2(c)).

5. Non-industrialized regions have greater opportunity for food loss and emissions prevention through optimized refrigeration than industrialized regions

The results of modeling food loss and waste, and emissions in the FSC from Agricultural Production through Consumption (though the impact of refrigeration was only assessed post farm-gate to retail) indicate that an optimized cold chain has significant potential to improve food losses and GHG emissions, particularly in non-industrialized regions (see figures 3(a) and (b)). Meanwhile, potential improvements are relatively modest in industrialized regions that already have highly developed cold chains. Key insights can be gained regarding regional improvement opportunities, the population density of the regions, and tradeoffs between food loss versus GHG emissions savings.

On an absolute basis, South and Southeast Asia and Sub-Saharan Africa have large food losses and possess some of the greatest opportunities for improvement (figure 3(a)). In South and Southeast Asia, over 83 billion kg of fruits & vegetables, 53 billion kg of cereals, and 48 billion kg of milk products can be saved through optimized refrigeration (figure 3(a)). In Sub-Saharan Africa, over 46 billion kg of meat, and 48 billion kg of root & tubers can be saved through optimized refrigeration (figure 3(a)). Meat losses dominate the results for GHG emissions associated with food loss, which is to be expected given the high GHG intensity of meat production. Nevertheless, figure 3(b) shows a striking opportunity for improvement. Sub-Saharan Africa's potential meat savings translate to a 700 MmtCO2-eq reduction (figure 3(b)). Meanwhile, fruits & vegetables, cereals, and roots & tubers in both regions produce few emissions relative to their high loss quantities (figure 3(b)).

Figures 3(c) and (d) illustrate the results when population is considered. While exhibiting some of the lowest absolute food loss and waste, and emissions, North America & Oceania has the most or second-most food loss and waste on a per capita basis across four of the six food types. Nevertheless, the overall improvement potential via cold chain optimization in North America & Oceania is low. Meanwhile, South & Southeast Asia has the largest absolute food losses, but lowest per capita food losses under current conditions. Despite these low per capita food losses, South & Southeast Asia has the potential to experience a 45% reduction in food losses and a 54% decrease in the associated emissions under an optimized refrigeration scenario. In contrast, Sub-Saharan Africa has the largest absolute and per capita food loss emissions, and tremendous opportunities for both food loss (47%) and emissions reduction (66%) under optimized refrigeration conditions.

Examining these results on a global basis, it is apparent that meat accounts for over 50% (2.7 Gt) of food loss and waste GHG emissions despite accounting for less than 10% (180 Mmt) of global food loss and waste (figures 4(a) and (b)). Optimized refrigeration of meat could result in the elimination of over 43% (1.1 Gt) of emissions associated with meat loss. Meanwhile, fruits & vegetables represent 30% of global food loss and waste but only 9% of GHG emissions. This relationship highlights a tradeoff between food loss prevention and GHG emissions mitigation and the importance of understanding this relationship when prioritizing food quantity or embodied emissions reduction.

6. Short supply chains have a greater effect on food losses than optimizing refrigeration quality

In practice, not all foods move through all stages of the supply chain shown in figure 2. Supply chains can be highly variable according to specific local conditions and specific kinds of food. To account for this variability, the extreme ends of long and short FSCs were modeled to assess potential differences between

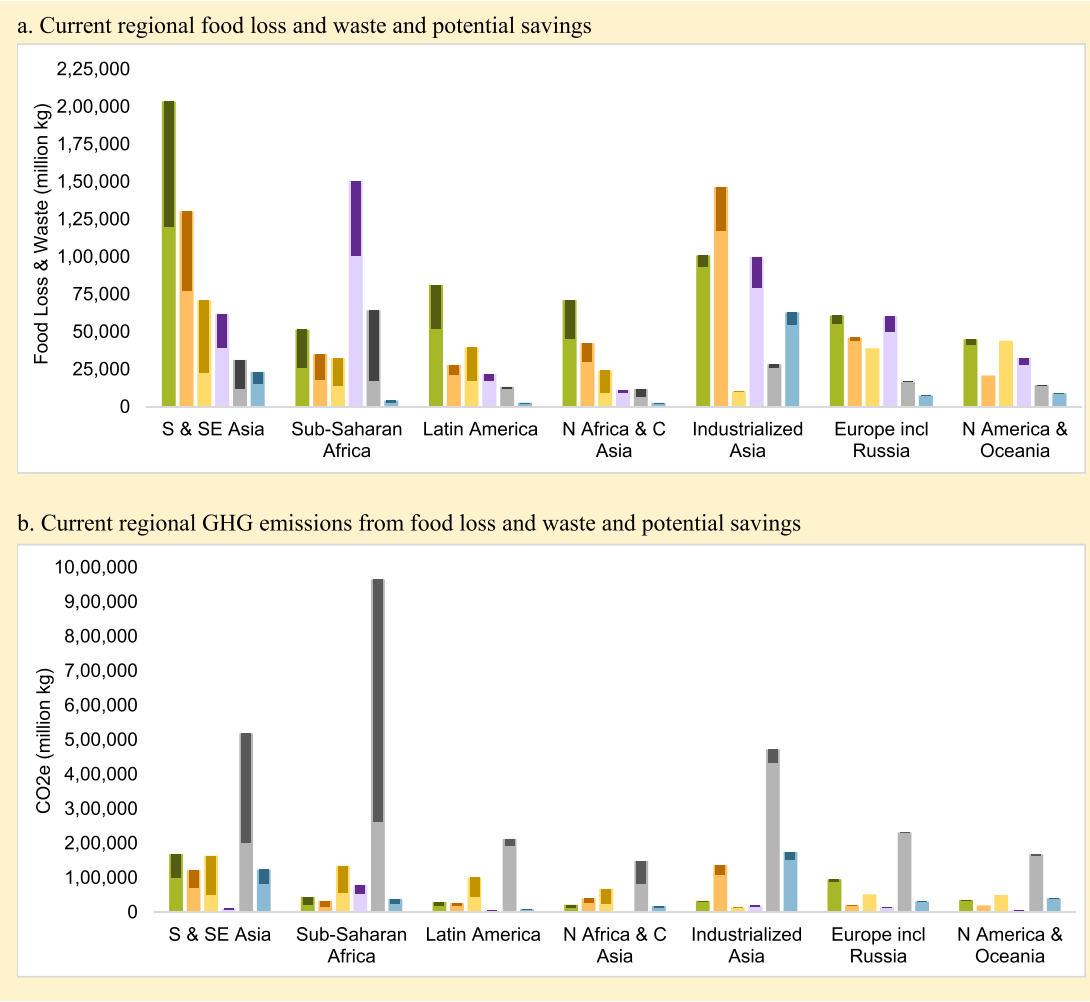


Figure 3. Regional food loss and waste, and associated emissions under current and optimally refrigerated FSC conditions.

hyper-localization and optimized industrial refrigeration. Figure 5 highlights three different scenarios: baseline (CC), optimized (OR), and short FSC (SF), representing the current case modeled in figures 2 and 3, the optimized case modeled in figure 3, and a highly localized, 'farm-to-table' food system without storage, processing, distribution, or the associated transportation stages, respectively.

Modeling these three scenarios showed that both short and longer optimized FSCs experience lower food losses compared with the baseline (figure 5). This pattern holds in non-industrialized and industrialized regions alike, though it is less pronounced in industrialized regions. This suggests that in developing contexts, improvements are possible by introducing optimized refrigeration or by making (or in many cases keeping) FSCs short. The actual feasibility of short FSCs is highly variable and dependent on geography, seasonality, and specific food type. Issues of food security and adequate nutrition, for example, are not addressed by these results. All FSCs experienced greater food loss savings from shortening the supply chain than from optimizing refrigeration. The disproportionate amount of on-farm and consumer losses in North America and Europe highlights the need for solutions addressing food loss and waste in higher income countries to focus more heavily on sources of loss outside of the actual supply chain. Despite this, it appears that shortened FSCs in industrialized regions can reduce food losses within the supply chain beyond what has already been accomplished through nearly optimal refrigeration.

7. Discussion

Although cold chain infrastructure is rapidly increasing, an optimized cold chain will likely develop at different rates and in different ways across the globe. This analysis demonstrates that while increased refrigeration should lead to improvements in both food loss and GHG emissions associated with food loss, there are important tradeoffs associated with cold chain improvements by food type and by region. Investment decisions will need to be prioritized to maximize the desired outcomes and impacts. As previously mentioned, improved food systems are aligned with a number of the Sustainability

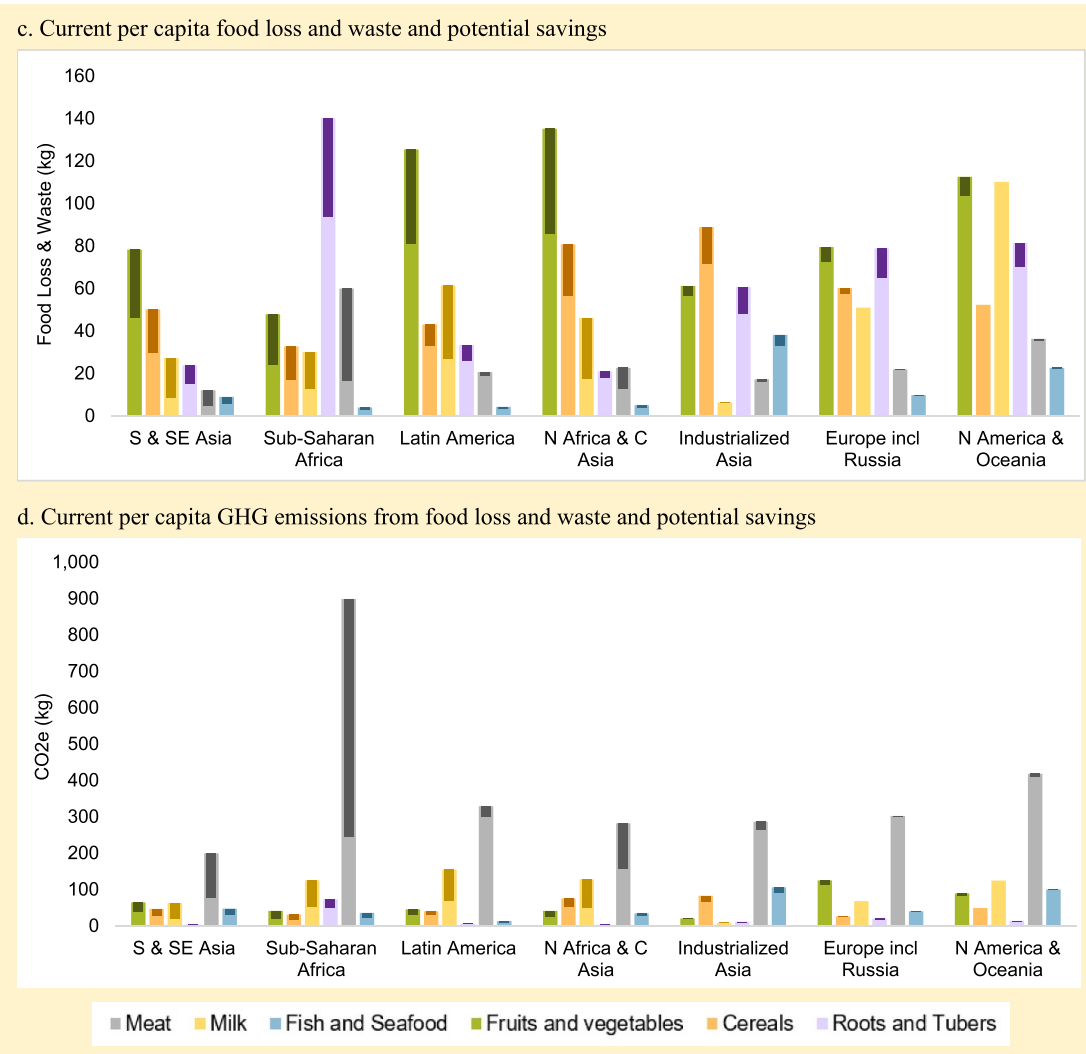


Figure 3. (Continued.)
Notes: Figure 3 shows current food loss and waste, the associated emissions, and the potential reduction opportunity of an optimized cold chain (darkened upper portions of the bars). Figures 3(a) and (b) show the total quantity of regional loss and waste, and the associated GHG emissions, respectively; (c), (d) show regional per capita loss and waste, and the associated GHG emissions, respectively. Oilseeds & pulses are excluded due to their small contribution.

Development Goals. If the SDG for Zero Hunger is the most important consideration, cold chain interventions that provide the greatest overall food loss reductions and best nutritional outcomes may best meet that objective. Alternatively, organizations that prioritize Climate Action may focus on reducing meat losses specifically rather than total food losses. In addition, considerations of total impact versus per capita impact have different patterns of improvement potentials.

The results indicate that Sub-Saharan Africa and South & Southeast Asia have the greatest overall potential for reductions in both food losses and emissions from increased cold chain implementation. Depending on the food type being targeted, food losses appear to experience the greatest reduction when refrigeration is implemented in the post-harvest handling & storage, transportation 2, and processing & packaging phases in non-industrialized regions. However, there are some inherent tradeoffs depending on whether improvements are targeted at reducing food loss or at reducing GHG emissions associated with food loss. As others have noted, enhanced cold chain infrastructure produces limited improvements in industrialized regions (Gustavsson *et al* 2011, FAO 2015, Heard and Miller 2019).

Additionally, non-industrialized and industrialized regions appear to experience food loss reductions from short FSCs beyond those from optimally refrigerated FSCs. While refrigeration helps reduce food degradation rates, reduced time within the overall supply chain can have a greater effect on food loss (Zanoni and Zavanella 2012, Kitinoja 2013, Abiso *et al* 2015). Both solutions, however, have their limitations. Short FSCs are often unable to supply adequate nutrition throughout the year due to the productivity of a region, the seasonality

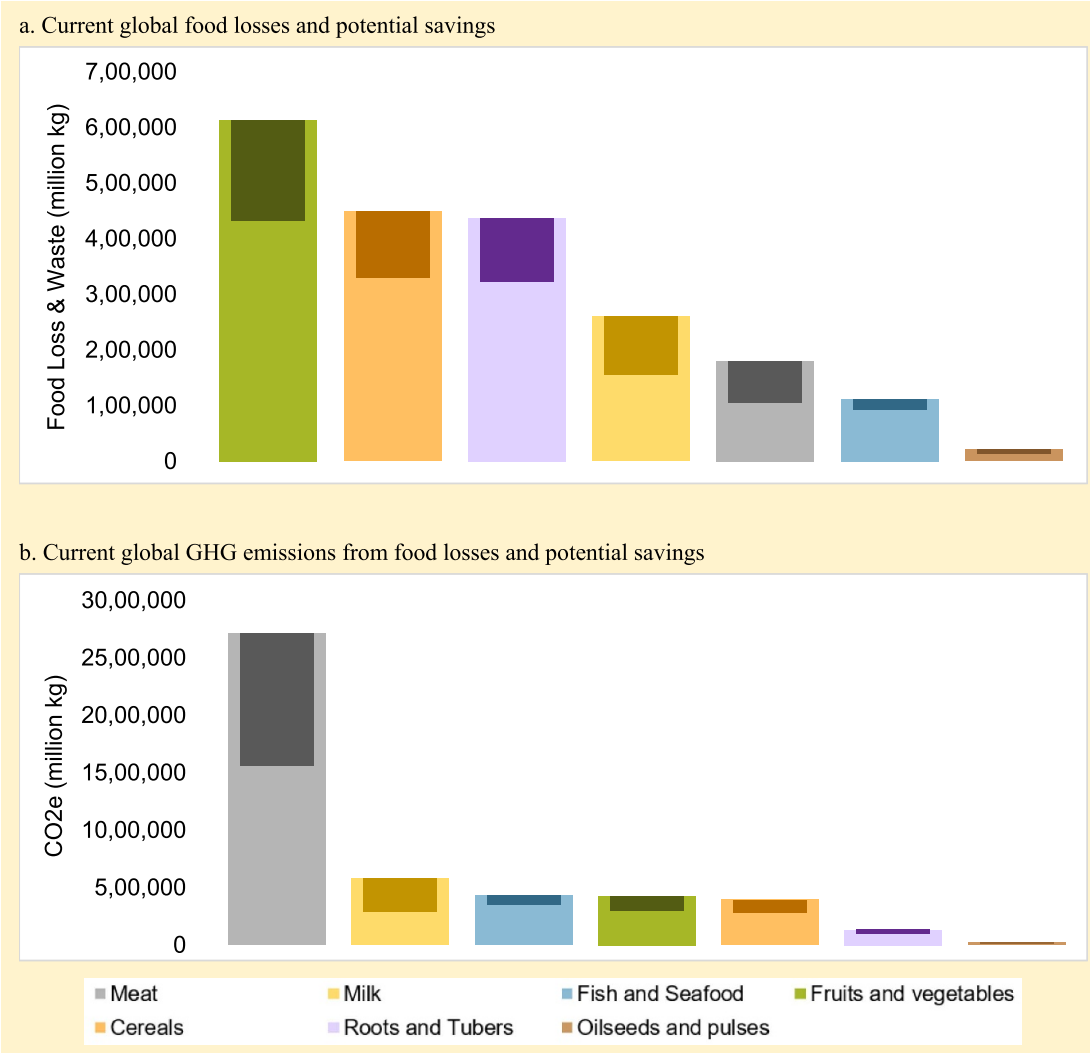


Figure 4. Global food losses and associated emissions under current and optimally refrigerated FSC conditions.
Notes: Figures 4(a) and (b) shows current food loss and waste (a) and the associated emissions (b), and the potential reduction opportunity of an optimized cold chain (darkened upper portions) globally.

of agriculture, or distance from a particular food resource (i.e. fisheries). Meanwhile, optimally refrigerated FSCs necessitate that significant infrastructural preconditions (energy, roads, logistics services) be met. This speaks to the need for nuanced, regionally appropriate solutions, especially as increasing climate variability continues to shift global food production and increasingly burden global infrastructure. An 'optimized' food system does not inherently mean highly globalized and industrialized for all products. While cold chain deployment can reduce food losses, it should accompany, rather than displace, robust, well-functioning localized food systems. By coupling these solutions, stakeholders can reduce food losses while simultaneously avoiding some of the energy burden and emissions impacts of refrigeration (Heard and Miller 2019, Hu *et al* 2019, International Institute of Refrigeration (IIR) 2021) and the aforementioned cultural loss and health risks.

This study provides novel insights on the potential improvements associated with cold chain introductions or upgrades; however, it has some obvious limitations. First, this study focuses solely on food losses and the associated emissions. It does not consider emissions associated with operating the cold chain nor any changes that could be induced in the system due to the presence of refrigeration, which researchers have found to produce a net increase in the overall emissions of the FSC (Heard and Miller 2019, Hu *et al* 2019, IIR 2021). For example, studies have shown that cold chain development can change community and regional diets resulting in food systems that are increasingly energy (and emissions) intensive and dependent upon refrigeration (Garnett 2011, Heard and Miller 2019). Further, reductions in food loss associated GHGs only reduce the non-productive GHG emissions associated with the food system, but does not necessarily decrease total GHG emissions. Improved supply chains could

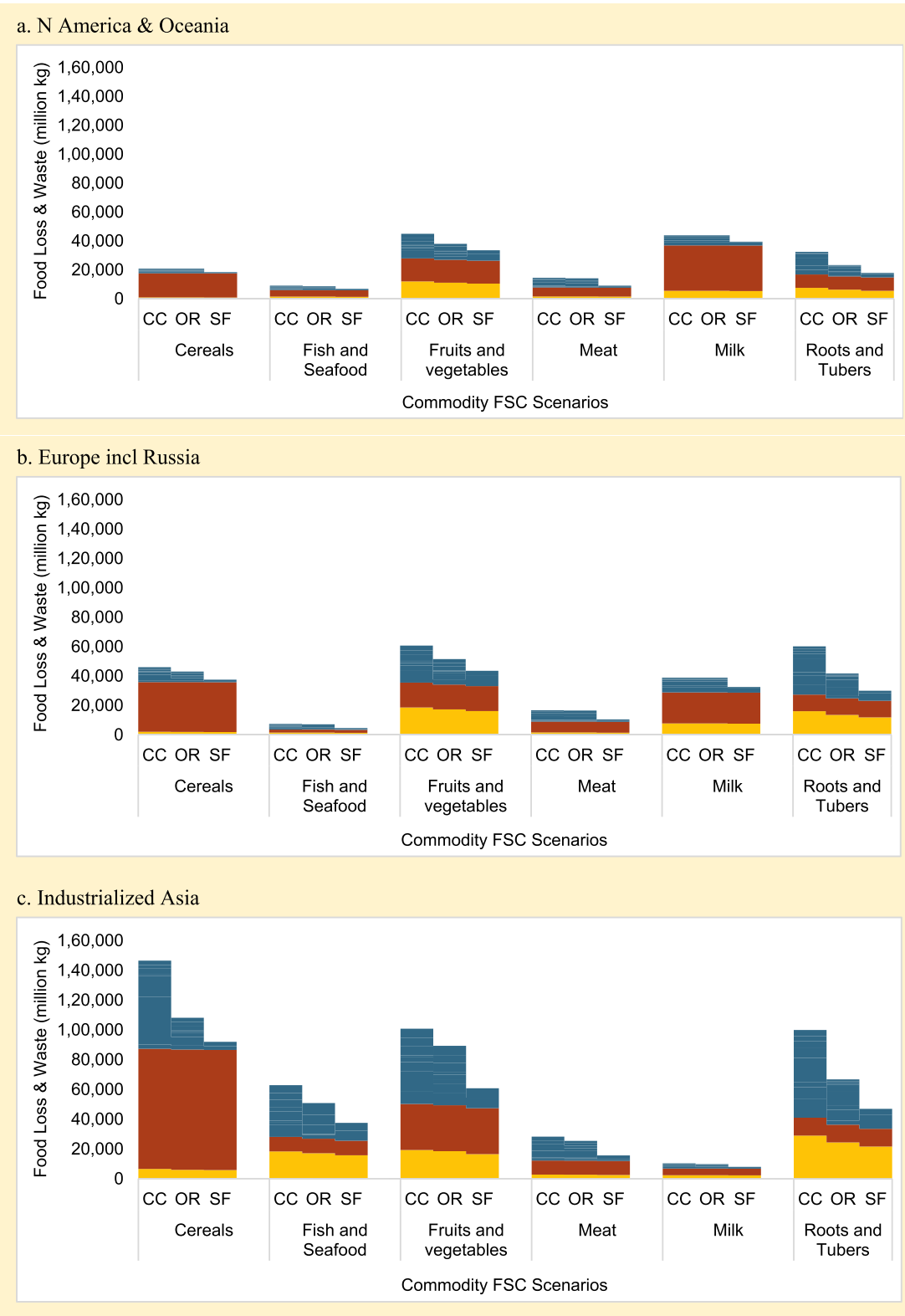


Figure 5. The relative differences of short and long supply chains on food loss.

lead to increased access and availability of food for human consumption, potentially redistributing food to address issues of global hunger. While this is a favorable outcome, it will not result in decreased agricultural production nor the associated GHGs.

An optimized cold chain may result in lower food loss, but does not consider a myriad of social, cultural, political, and economic factors that shape a food system. The study does not consider nutritional qualities of different food types or the social and

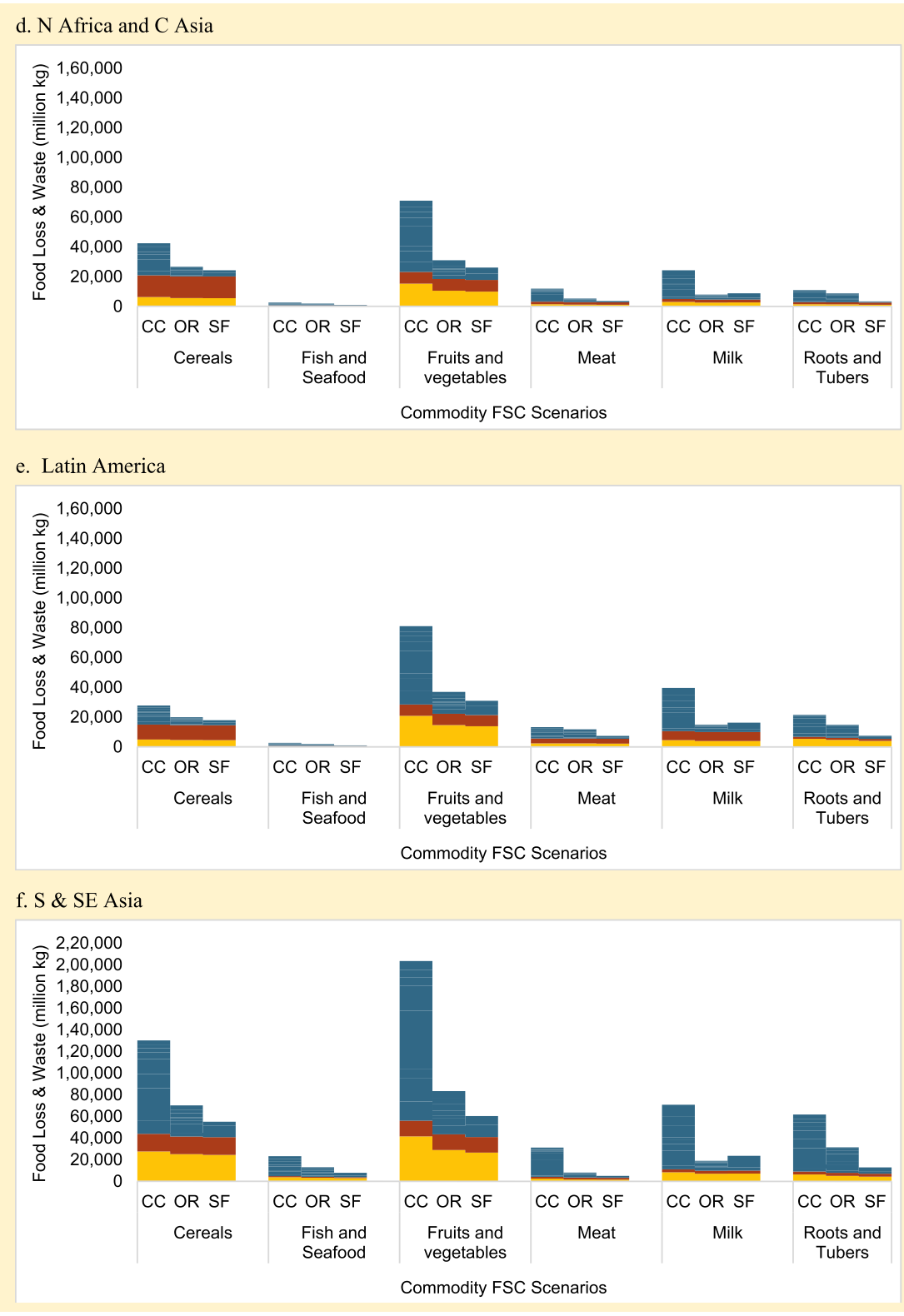


Figure 5. (Continued.)

cultural importance of food. From a technology standpoint, reliable energy infrastructure is a baseline requirement for cold chain expansion to improve access to nutrition and reduce foodborne illnesses (Aung and Chang 2014, Mercier *et al* 2018). However, in regions that lack the underlying infrastructure necessary for an effective cold chain, an ineffective cold chain coupled with a diet dependent upon cold chain infrastructure could result in greater food loss, food insecurity, and emissions while potentially weakening cultural heritage and self-sufficiency (Mercier *et al* 2017, UNEP and FAO 2022).

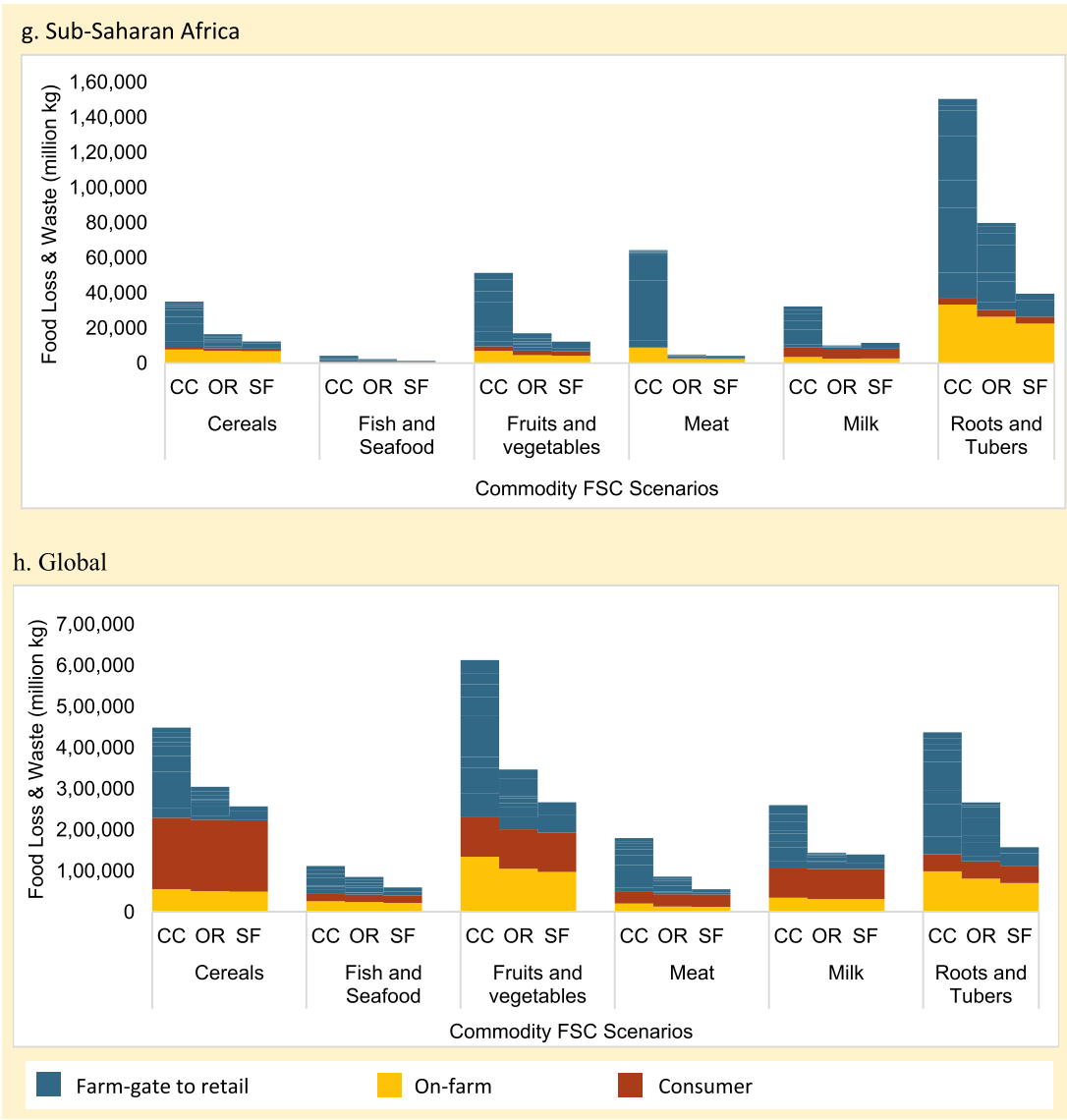


Figure 5. (Continued.)
Notes: Figure 5: Food loss and waste for six food types are modeled under three conditions: baseline (CC), optimized (OR), and short FSC (SF) (illustrated in figure 1(b)) regionally (a)–(g) and globally (h). While food loss and waste in panels (a)–(e) and (g) is represented on equal scales, food loss and waste in S&SE Asia (f) and globally (h) are recorded on larger scales. Unlike the previous figures, figures 5 highlights both the relative contributions of food losses within the supply chain (i.e. farm-gate to retail) and those typically considered outside the supply chain. Loss and waste at all stages were calculated based on a constant state of demand (consumption), thus as food losses decrease in the farm-gate to retail stages, on-farm production loss and waste decrease. The variability in these changes is due to different regional production and consumption stage loss rates.

As with air conditioning, the irony of refrigeration is that it will increasingly become a necessary tool of our FSCs as climate change worsens. Thus, as non-industrialized regions continue growing technologically and in population, it is critical to ensure that any technology deployed in these regions is implemented sustainably and in a manner that increases community resilience.

Although the analysis presented can be used to identify major trends and opportunities across regions and food types, the underlying data on actual food loss rates remain uncertain and variability can exist within regions. Projections of food losses based on historical trends do not appear to align well with theoretical food degradation models (Dong and Miller 2021, FAOSTAT 2023, Hu *et al* 2019, Wu *et al* 2019, Zanoni and Zavanella 2012). This gap could result from many factors; for instance, others have noted that the FAO's food loss and waste data are limited and in many cases inconsistent and uncertain due to evolving definitions, varying tracking and reporting methodologies, and data access and quality limitations (Chaboud and Daviron 2017, Lipińska *et al* 2019, Parfitt *et al* 2010, Xue *et al* 2017). Additionally, food loss estimates could further be improved by incorporating a quality degradation factor to account for potential downstream FSC losses caused by sub-optimal upstream conditions. Future work on this

topic can and should account for nutritional aspects of food (i.e. calories, protein, micronutrients) instead of just total mass. While it does not directly address that gap, by adding greater flexibility to FSC models, this study provides a new way to probe the discrepancy between methodological approaches.

While the results of this study align with the results of previous studies (Gustavsson *et al* 2011, FAO 2019, IIR 2020) relative to the percent food loss, savings opportunities, and ratios of food types within the supply chain, the quantities associated with these percentages are significantly greater in this study than in previous studies. This difference is primarily a result of methodological differences between this study and prior studies. Most previous studies utilize conversion factors to calculate the edible quantity of food produced. For example, Gustavsson *et al* (2011) use a 50% conversion factor for fish and seafood, meaning that only 50% of fish and seafood produced is accounted for as food. Since a key value of this research and the model is quantifying GHG emissions associated with food loss, a conversion factor was not used. This methodology aligns with Porter *et al* (2016) who cite the importance of accounting for 'the entire food commodity', as any resulting loss has embedded emissions. Additionally, a small fraction of the difference in food production results in the current study reflects the increase in food produced globally over time—this research used the most recent (2020) FAO Food Balance data in contrast to that of 2009 (Gustavsson *et al* 2011) and 2016 (IIR 2020). Between 2019 and 2020 alone, the FAO reported ∼500 Gt more food produced (FAOSTAT 2023).

The findings of this study can be utilized by various FSC stakeholders. Farmers, food logistics firms, and food retailers can use this model to optimally utilize cold chain technologies to better service their customers. International NGOs and intergovernmental bodies can use it to deploy resources targeted at reducing food loss, hunger, and climate change. For example, recognizing the role of sustainable cold chain infrastructure, Germany led the UN's Green Cooling Initiative (GCI) in 2020, which aims to reduce emissions from the cooling sectors by prioritizing sustainable refrigerants, energy efficiency, and energy consumption (Deutsche Gesellschaft für Internationale Zusammenarbeit n.d., UN 2012). This research provides a critical supplement to GCI as it answers the question of where within a given FSC, and at what intensity, refrigeration can be deployed most effectively.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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